

Manogya Dahal

him@manogyadahal.com.np | manogyadahal.com.np | github/ManogyaDahal | +977-9761795280

A Computer Science undergraduate with experience working with servers, systems architecture, and open-source development. Proficient in backend projects across multiple programming languages.

Education

Kathmandu University, Dhulikhel, Kavrepalanchowk

Aug. 2023 – Exp. Aug. 2027

BSc in Computer Science

Relevant Coursework: Algorithms, Databases, Computer Networks, Data Structures, Operating Systems

Projects

Collaboratex: (Go, React, Redis, Docker)

- Engineered a concurrent compilation pipeline utilizing Go and Redis caching to efficiently manage multi-user document build requests.
- Built real-time collaborative text-editor and chat system using codemirror and gorilla websockets.

Letsgodb (Go)

- Designed and implemented a SQL-like database engine from scratch.
- Built a custom query parser, file-based storage layer, and CRUD execution engine.

DistQ (Go, Javascript, Redis)

- Worked on the broker and queue part of the task queue.
- Designed the dashboard displaying the status of the task and workers

GoType (Go, React)

- Engineered a real-time multiplayer synchronization engine utilizing WebSockets.
- Setup google's oauth along with chat system inside lobby and player's progress

Concurrent-grep (Go)

- Reimplemented the grep utility in go with concurrency
- Recursively searches for the match inside of the directories and sub-directories

Http/1.1 (Go)

- Engineered a custom HTTP/1.1 server from scratch utilizing net package, managing raw TCP connections.
- Designed and built a byte-stream request parser to decode http protocol specifications, extracting request lines, parsing headers, and managing status code routing.

Conways game of life (Go, Ebitengine)

- Engineered a 2D cellular automata engine using the Ebiten game engine in Go, implementing custom game loops for real-time grid rendering and state updates.

BMP (Python, Opencv)

- Engineered a real-time gesture-controlled media player utilizing OpenCV to track hand movements and calculate precise finger counts for UI navigation.

Technical Skills

- Languages:** Go, Python, C/C++, Rust, TypeScript, Bash, Lua
- Frameworks/Libraries:** React, Django, Node.js, Pygame, Tailwind, Gin, Sevelte
- Other:** Vim, Git, Linux, Docker & Virtualization, ORMs, Sockets, REST APIs
- Databases/Caching:** MySQL, SQLite, PostgreSQL, MongoDB, Redis

Experience

Technician – Kathmandu University High Performance Computing (KUHPC) Jun 2024 – Aug 2025

- Assisted in maintaining and troubleshooting the university's HPC cluster.
- Monitored and optimized system performance, handled system updates and hardware maintenance.

Leadership & Activities

Active contributor – Kathmandu University Open Source Community (KUOSC)	<i>Aug 2024 – Present</i>
Active member – Kathmandu University Computer Club (KUCC)	<i>2023 - Present</i>